DAA LAB — WEEK 2
Programs 6–10
Source: extracted from your uploaded DAA lab PDF.

PROGRAM 6: Minimum number of Indian currency notes/coins

```
#include <stdio.h>
int main() {
   int V, count = 0;
   int denominations[] = {1000, 500, 100, 50, 20, 10, 5, 2, 1};
   scanf("%d", &V);
   for (int i = 0; i < 9; i++) {
       count += V / denominations[i];
       V %= denominations[i];
   }
   printf("%d", count);
   return 0;
}
```

PROGRAM 7: Assign cookies to maximize content children

```
#include <stdio.h>
#include <stdlib.h>
int compare(const void *a, const void *b) {
   return (*(int *)a > *(int *)b) - (*(int *)a < *(int *)b);
}
int main() {
   int n, m;
   scanf("%d", &n);
   int g[n];
   for (int i = 0; i < n; i++)
       scanf("%d", &g[i]);
   scanf("%d", &m);
   int s[m];
   for (int i = 0; i < m; i++)
       scanf("%d", &s[i]);
   qsort(g, n, sizeof(int), compare);
   qsort(s, m, sizeof(int), compare);
   int i = 0, j = 0, count = 0;
   while (i < n && j < m) {
       if (s[j] >= g[i]) {
           count++;
           i++;
       }
       j++;
   }
   printf("%d", count);
   return 0;
```

```
}
```

```
================================================================================
PROGRAM 8: Minimum calories running distance (greedy)
================================================================================
```

```
#include <stdio.h>
#include <math.h>
int main() {
   int n;
   scanf("%d", &n);
   int arr[n];
   for (int i = 0; i < n; i++) {
       scanf("%d", &arr[i]);
   }
   for (int i = 0; i < n - 1; i++) {
       for (int j = 0; j < n - i - 1; j++) {
           if (arr[j] < arr[j + 1]) {
               int temp = arr[j];
               arr[j] = arr[j + 1];
               arr[j + 1] = temp;
           }
       }

   }
   int s = 0;
   for (int i = 0; i < n; i++) {
       s = s + (pow(n, i) * arr[i]);
   }
   printf("%d", s);
   return 0;
}
```

```
================================================================================
PROGRAM 9: Maximize sum of arr[i] * i
================================================================================
```

```
#include <stdio.h>
#include <stdlib.h>
int compare(const void *a, const void *b) {
   return (*(int *)a - *(int *)b);
}
int main() {
   int n;
   scanf("%d", &n);
   int arr[n];

   for (int i = 0; i < n; i++)
       scanf("%d", &arr[i]);
   qsort(arr, n, sizeof(int), compare);
   long long sum = 0;
   for (int i = 0; i < n; i++)
```

```
        sum += (long long)arr[i] * i;
    printf("%lld", sum);
    return 0;
}
```

==========

PROGRAM 10: Minimum sum of products of two arrays

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```
#include <stdio.h>
#include <stdlib.h>
int ascending(const void *a, const void *b) {
    return (*(int *)a - *(int *)b);
}
int descending(const void *a, const void *b) {
    return (*(int *)b - *(int *)a);
}
int main() {
    int n;
    scanf("%d", &n);
    int a[n], b[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);
    for (int i = 0; i < n; i++)
        scanf("%d", &b[i]);

    qsort(a, n, sizeof(int), ascending);
    qsort(b, n, sizeof(int), descending);
    long long sum = 0;
    for (int i = 0; i < n; i++)
        sum += (long long)a[i] * b[i];
    printf("%lld", sum);
    return 0;
}
```